

• EXACT DISTRIBUTION of $\hat{\beta}_1$ and $\hat{\beta}_2$

Preliminary result

Given $Y_1, \dots, Y_n$ independent with distribution $Y_i \sim N(\mu_i, \sigma^2)$ $i=1, \dots, n$
and a sequence of known constants a_i , $i=1, \dots, n$,
 $\sum_{i=1}^n a_i Y_i \sim N\left(\sum_{i=1}^n a_i \mu_i, \sigma^2 \sum_{i=1}^n a_i^2\right)$

We have seen that $\hat{\beta}_1$ and $\hat{\beta}_2$ are linear combinations of $Y_1, \dots, Y_n$ of the form

$$\hat{\beta}_1 = \sum_{i=1}^n v_i Y_i \quad \hat{\beta}_2 = \sum_{i=1}^n w_i Y_i$$

hence $\hat{\beta}_1$ and $\hat{\beta}_2$ are exactly Gaussian-distributed r.v. (see res. 1)

Moreover, the expression of the two estimators are the same we obtained with OLS, In fact, the Gaussian error model is a special case, hence the properties we computed still hold.

In particular, we computed

$$E[\hat{\beta}_1] = \beta_1 \quad \text{var}(\hat{\beta}_1) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right)$$

$$E[\hat{\beta}_2] = \beta_2 \quad \text{var}(\hat{\beta}_2) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

The exact distributions are then easily obtained as

$$\hat{\beta}_1 \sim N\left(\beta_1, \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right)\right)$$

$$\hat{\beta}_2 \sim N\left(\beta_2, \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}\right)$$

• EXACT DISTRIBUTION of $\hat{\Sigma}^2$

$$\hat{\Sigma}^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{\beta}_1 - \hat{\beta}_2 x_i)^2$$

it is possible to show that

$$\frac{n \hat{\Sigma}^2}{\sigma^2} \sim \chi_{n-2}^2 \quad \text{Chi-squared with } n-2 \text{ degrees of freedom.}$$

In general, for a χ_ν^2 r.v., the expected value is ν

$$E\left[\frac{n \hat{\Sigma}^2}{\sigma^2}\right] = (n-2) \Rightarrow E[\hat{\Sigma}^2] = \frac{(n-2)}{n} \sigma^2 \quad \text{biased}$$

hence again we obtain an unbiased estimator as

$$S^2 = \frac{n}{n-2} \hat{\Sigma}^2 \quad E[S^2] = \frac{n}{n-2} E[\hat{\Sigma}^2] = \frac{n}{n-2} \cdot \frac{n-2}{n} \sigma^2 = \sigma^2$$

$$\text{and } \frac{(n-2) S^2}{\sigma^2} \sim \chi_{n-2}^2.$$

Moreover, it is possible to show that $\hat{\Sigma}^2 \perp (\hat{\beta}_1, \hat{\beta}_2)$
(hence, also $S^2 \perp (\hat{\beta}_1, \hat{\beta}_2)$)

INFERENCE ABOUT β

We have derived the exact distributions of the estimators.

With these distributions we can test statistical hypotheses, compute confidence intervals.

Examples

$$\text{Test: } \begin{cases} H_0: \beta_j = b \\ H_1: \beta_j \neq b \end{cases} \quad \begin{cases} H_0: \beta_j = 0 \\ H_1: \beta_j > 0 \end{cases} \quad j=1,2$$

$$\text{Confidence interval, of level } 1-\alpha, \quad \hat{C}_j \text{ such that } P(\hat{C}_j \ni \beta_j) = 1-\alpha \quad \forall \beta_j \in \mathbb{R}$$

$$\text{Recall that: } \hat{\beta}_1 \sim N(\beta_1, V(\hat{\beta}_1))$$

$$\hat{\beta}_2 \sim N(\beta_2, V(\hat{\beta}_2))$$

$$\frac{(n-2) S^2}{\sigma^2} \sim \chi_{n-2}^2$$

$$\text{where } V(\hat{\beta}_1) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right)$$

$$V(\hat{\beta}_2) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

• CONFIDENCE INTERVAL for β_j

Preliminary result:

If $Z \sim N(0,1)$ and $W \sim \chi_\nu^2$ independent, then $\frac{Z}{\sqrt{W/\nu}} \sim t_\nu$.

(Student's t with ν degrees of freedom)

↳ symmetric distrib.

• heavier tails than a normal

• for large ν it is very close to a normal

First of all, we need to find a pivotal quantity.

PIVOTAL QUANTITY: a transformation of the random variables (and of the parameter) whose distribution does not depend on the parameter (hence is completely known).

Since $\hat{\beta}_j \sim N(\beta_j, V(\hat{\beta}_j))$, the simplest (and most intuitive) transformation is

$$\frac{\hat{\beta}_j - \beta_j}{\sqrt{V(\hat{\beta}_j)}} \sim N(0,1)$$

however, $V(\hat{\beta}_j)$ includes σ^2 which is unknown

In place of $V(\hat{\beta}_j)$ we use an estimate, $\hat{V}(\hat{\beta}_j) = \frac{S^2}{\sigma^2} V(\hat{\beta}_j)$ (eq. $\hat{V}(\hat{\beta}_2) = \frac{S^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$)

$$T_j = \frac{\hat{\beta}_j - \beta_j}{\sqrt{\hat{V}(\hat{\beta}_j)}}$$

hence random!

what is its distribution? Notice that $\hat{V}(\hat{\beta}_j)$ includes $\hat{\Sigma}^2$ (transformation of Y)

$$T_j = \frac{\hat{\beta}_j - \beta_j}{\sqrt{\hat{V}(\hat{\beta}_j)}} = \frac{\frac{\hat{\beta}_j - \beta_j}{\sqrt{V(\hat{\beta}_j)}}}{\sqrt{\frac{\hat{V}(\hat{\beta}_j)}{V(\hat{\beta}_j)}}} \rightarrow \hat{V}(\hat{\beta}_j) = \frac{S^2}{\sigma^2} V(\hat{\beta}_j) \Rightarrow \frac{\hat{V}(\hat{\beta}_j)}{V(\hat{\beta}_j)} = \frac{S^2}{\sigma^2} \cancel{V(\hat{\beta}_j)} \cdot \frac{1}{\cancel{V(\hat{\beta}_j)}}$$

$$= \frac{\hat{\beta}_j - \beta_j}{\sqrt{V(\hat{\beta}_j)}} \cdot \frac{1}{\sqrt{\frac{S^2}{\sigma^2} \cdot \frac{(n-2)}{(n-2)}}}$$

$$= \frac{\frac{\hat{\beta}_j - \beta_j}{\sqrt{V(\hat{\beta}_j)}} \sim N(0,1)}{\sqrt{\frac{(n-2) S^2}{\sigma^2} \cdot \frac{1}{(n-2)}}} = \sqrt{\frac{\chi_{n-2}^2}{n-2}}$$

moreover, $\hat{\beta}_j \perp S^2$

$$\Rightarrow T_j \sim t_{n-2}$$

CONFIDENCE INTERVAL: We want to find an interval (u, v) such that

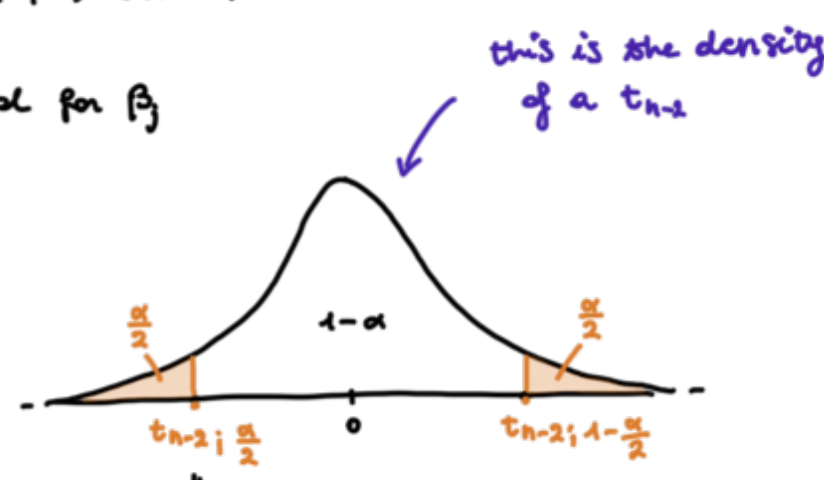
$$P(u < T_j < v) = 1-\alpha$$

and then "isolate" the parameter to find an interval for β_j

$$T_j \sim t_{n-2} \quad \text{hence}$$

$$P\left(t_{n-2; 1-\frac{\alpha}{2}} < T_j < t_{n-2; 1-\frac{\alpha}{2}}\right) = 1-\alpha$$

quantile $1-\frac{\alpha}{2}$ of a t_{n-2} distrib.



$$P\left(-t_{n-2; 1-\frac{\alpha}{2}} < \frac{\hat{\beta}_j - \beta_j}{\sqrt{\hat{V}(\hat{\beta}_j)}} < t_{n-2; 1-\frac{\alpha}{2}}\right) = 1-\alpha$$

$$P\left(\hat{\beta}_j - \sqrt{\hat{V}(\hat{\beta}_j)} \cdot t_{n-2; 1-\frac{\alpha}{2}} < \beta_j < \hat{\beta}_j + \sqrt{\hat{V}(\hat{\beta}_j)} \cdot t_{n-2; 1-\frac{\alpha}{2}}\right) = 1-\alpha$$

$$P(\beta_j \in \hat{C}) = 1-\alpha \quad \text{where } \hat{C} = \hat{\beta}_j \pm \sqrt{\hat{V}(\hat{\beta}_j)} \cdot t_{n-2; 1-\frac{\alpha}{2}}$$

$\hat{C}$ is a random interval. After observing the data we can compute its realization

by substituting the estimators with their estimates.

$$\text{We obtain } \beta_j \in \hat{\beta}_j \pm t_{n-2; 1-\frac{\alpha}{2}} \sqrt{\hat{V}(\hat{\beta}_j)}.$$

$$\text{That is } \beta_1 \in \hat{\beta}_1 \pm t_{n-2; 1-\frac{\alpha}{2}} \sqrt{\frac{S^2}{\sigma^2} \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right)}$$

$$\beta_2 \in \hat{\beta}_2 \pm t_{n-2; 1-\frac{\alpha}{2}} \sqrt{\frac{S^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

• HYPOTHESIS TEST on β_j

$$\begin{cases} H_0: \beta_j = b \\ H_1: \beta_j \neq b \end{cases}$$

"under H_0 " means: assuming H_0 true, we study the distribution of the test statistic assuming H_0 true, and study which values point against it.

Following the same reasoning as before, we use the TEST STATISTIC

$$T_j = \frac{\hat{\beta}_j - b}{\sqrt{\hat{V}(\hat{\beta}_j)}} \stackrel{H_0}{\sim} t_{n-2} \quad \text{(under } H_0 \text{ } b \text{ is the true value of the param, so we are subtracting the true mean of } \beta_j \text{)}$$

T_j is a random variable. After observing $Y_1, \dots, Y_n$ we can compute the OBSERVED VALUE OF THE TEST

$$t_{\text{obs}} = \frac{\hat{\beta}_j - b}{\sqrt{\hat{V}(\hat{\beta}_j)}} \quad \text{we substitute the estimates: not random! it is a number}$$

How do we define the acceptance and reject regions?

We ask: "what values of the test do we expect when H_0 is true? And what values do we expect instead when H_0 is not true (under H_1)?"

If H_0 true: $\beta_j = b$.

If the data support this hypothesis, then the estimate $\hat{\beta}_j$ will be close to b ($E[\hat{\beta}_j] = \beta_j = b$).

$$\Rightarrow \hat{\beta}_j - b \approx 0 \Rightarrow t_{j, \text{obs}} \approx 0$$

Hence we expect that, if H_0 is true, t_{obs} will be small (in absolute value)

If H_0 is not true, then $\beta_j \neq b$. The estimate $\hat{\beta}_j$ will be different from b

$$\Rightarrow |\hat{\beta}_j - b| \text{ large} \Rightarrow |t_{j, \text{obs}}| \text{ large}$$

Hence we expect that, under H_1 , $t_{j, \text{obs}}$ will be large (in absolute value)

The acceptance region thus will contain the values around 0 $(-a, +a) = A$

The reject region will contain values far from 0 $(-\infty, -a) \cup (a, +\infty) = R$

We need to define the thresholds $-a, a$

$$(A) \text{ fixed significance level } \alpha: \alpha = P(\text{reject } H_0 \mid H_0 \text{ true}) = P_{H_0}(T_j \in R) = P_{H_0}((T_j < -a) \cup (T_j > +a))$$

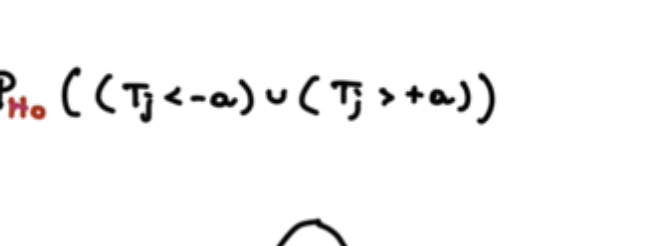
$$P_{H_0}(|T_j| > t_{n-2; 1-\frac{\alpha}{2}}) = \alpha$$

the acceptance region is $A = (t_{n-2; 1-\frac{\alpha}{2}}, t_{n-2; 1-\frac{\alpha}{2}})$

the reject region is $R = R_1 \cup R_2 = (-\infty; t_{n-2; 1-\frac{\alpha}{2}}) \cup (t_{n-2; 1-\frac{\alpha}{2}}; +\infty)$

if $t_{j, \text{obs}} \in A \Rightarrow$ we do not reject H_0

if $t_{j, \text{obs}} \notin A \Rightarrow$ we reject H_0



(B) p-value

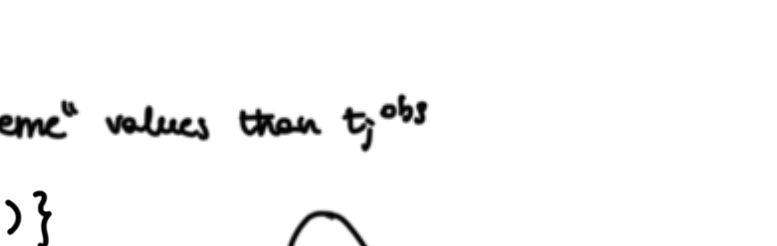
it is the probability of observing "more extreme" values than $t_{j, \text{obs}}$

$$\alpha^{\text{obs}} = 2 \min\{P_{H_0}(T \geq t_{j, \text{obs}}); P_{H_0}(T \leq t_{j, \text{obs}})\}$$

the t distribution is symmetric, so

$$\alpha^{\text{obs}} = P_{H_0}(|T_j| > |t_{j, \text{obs}}|)$$

$$= 2 \cdot P_{H_0}(T_j > |t_{j, \text{obs}}|)$$



connection between the two types of test

- if $\alpha^{\text{obs}} < \alpha \Rightarrow$ reject H_0 at level α

- if $\alpha^{\text{obs}} > \alpha \Rightarrow$ do not reject H_0 at a level α

In practical applications, these methods are useful tools to investigate relevant applicative questions. For example:

• does the covariate x have a significant effect on Y ?

The effect of x on Y is summarised by the coefficient β_1 .

Hence this question can be formalised by the statistical test

$$\begin{cases} H_0: \beta_1 = 0 \rightarrow \text{no effect} \\ H_1: \beta_1 \neq 0 \end{cases}$$

Indeed the model $Y_i = \beta_1 + \beta_2 x_i + \epsilon_i$

under H_0 becomes $Y_i = \beta_1 + \epsilon_i$ (x has no impact on Y)

↳ this is called the "NULL MODEL"